

Answer: D

13 On dry ground,

$$F = \frac{mv^2}{r}$$

$$v = \sqrt{\frac{Fr}{m}} = 16 \text{ m s}^{-1}$$

On wet ground,

$$\frac{F}{2} = \frac{mv_1^2}{r}$$

$$v_1 = \sqrt{\frac{Fr}{2m}} = \frac{16}{\sqrt{2}} \text{ m s}^{-1}$$

Answer: D